Exercise Sheet 7

Structural Balance

5942UE Network Science

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7.A Triadic Balance Theory

Exercise 7.A.1: Classifying Signed Triangles

Label each signed triangle as **balanced (B)** or **unbalanced (U)**:

1. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{+}{-} C$ (all positive)
2. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{-}{-} C$ (two positive, one negative)
3. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{+}{-} C$ (one positive, two negative)
4. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{-}{-} C$ (all negative)

Which pattern is most stable, and which is "socially unstable"?

Exercise 7.A.2: Applying the Balance Theorem

Complete signed graph on $1, 2, 3, 4, 5$. All edges positive **except** 1–3, 1–4, 2–3, 2–4 (which are negative).

1. Check whether every triangle is balanced.
2. If balanced, find the **two-camp partition** guaranteed by the Balance Theorem.
3. Verify within-camp edges are positive and between-camp edges negative.
4. What does the partition represent socially?

7.B Weak Balance

Exercise 7.B.1: Strong vs. Weak Balance

Graph with two triangles connected by one edge:

- A, B, C all positive
- D, E, F all negative
- Connecting edge C–D is negative

1. Is A, B, C balanced under strong balance? Under weak balance?
2. Is D, E, F balanced under strong balance? Under weak balance?
3. What does the global partition look like under each?
4. In international relations, what does an all-negative triangle represent?

Exercise 7.B.2: Finding Camps in a Signed Network

A signed graph: $1, 2, 3$ are mutually friends, $4, 5, 6$ are mutually friends, all cross-group edges are negative.

1. Check every triangle for balance.
2. Find the two-camp partition.
3. Verify within-group edges are positive, between-group negative.
4. Add a "rogue" positive edge $1 \overset{+}{-} 4$. Which triangles become unbalanced?

7.C Relaxed Balance and Applications

Exercise 7.C.1: WWI Alliance Network

Five countries during WWI:

- **Allies**: France, Britain, Russia (mutually positive)
- **Central Powers**: Germany, Austria-Hungary (mutually positive)
- All Allies–Central Powers edges are negative

1. Is this network perfectly balanced? Identify any unbalanced triangles.
2. Does it match the two-camp Balance Theorem?
3. What if a new country joins with positive ties to **both** camps?
4. Italy switched from Germany to the Allies in 1915 — what does balance theory predict?

Exercise 7.C.2: Approximate Balance in Real Networks

Using the karate club graph, sign edges by faction (within = +, cross = −).

1. Count balanced vs. unbalanced triangles.
2. What fraction of triangles are balanced?
3. Remove cross-faction edges in order of how many unbalanced triangles they participate in. After removing the top 5, how does the fraction change?
4. Plot fraction-balanced vs. edges-removed.

Wrap-Up

- triangle sign products give a one-line balance test
- the Balance Theorem turns a local rule into a global two-camp partition
- weak balance permits more than two camps and is often more realistic
- real networks are rarely perfectly balanced — approximate balance is the typical regime